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Software Requirement Specification (SRS)

PSB Hackathon 2026

Software Requirement Specification
Version 1

Project Name: Quantum Shield

TeamName: QCiphers

InstituteName: GMR INSTITUTE OF TECHNOLOGY

Revision History

Version No	Date	Prepared by	Significant Changes
Draft V1.0	15 Mar 2026	Abdul Yunus	Abdul Yunus

Declaration

The purpose of this Software Requirements Specification (SRS) document is to identify and document the user requirements for the **QuantumShield** The end deliverable software that will be supplied by **QCiphers** will comprise of all the requirements documented in the current document and will be operated in the manner specified in the document. The Source code will be developed subsequently based on these requirements and will formally go through code review during testing process.

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15 Mar 2026
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Software Requirement Specification (SRS)

Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to identify and document the user requirement for QuantumShield. This document is prepared with the following objectives:

- To provide behaviour of the system
- To provide Process Flow charts

1.2 Scope

- Discover cryptographic inventory (TLS certificates, VPN endpoints, APIs).
- Identify cryptographic controls (cipher suites, key exchange mechanisms, TLS versions).
- Validate whether deployed algorithms are quantum-safe.
- Generate actionable recommendations for non-PQC ready assets.
- Issue digital labels: Quantum-Safe, PQC Ready, or Fully Quantum Safe.
- Enterprise wide console for Central management: A GUI console to display status of scanned systems (public facing applications) covering details mentioned in Appendix-A (Cert-In CBOM Elements). As per the variation of score (like High, Medium, Low rating etc) for any public applications, dashboard should display that change as well.

1.3 Intended Audience

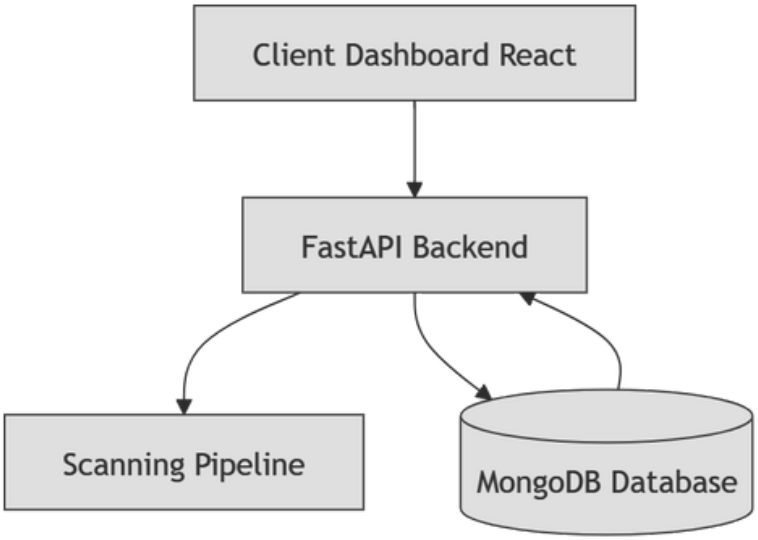
The intended audience of this document is only users of PNB.

Overall Description

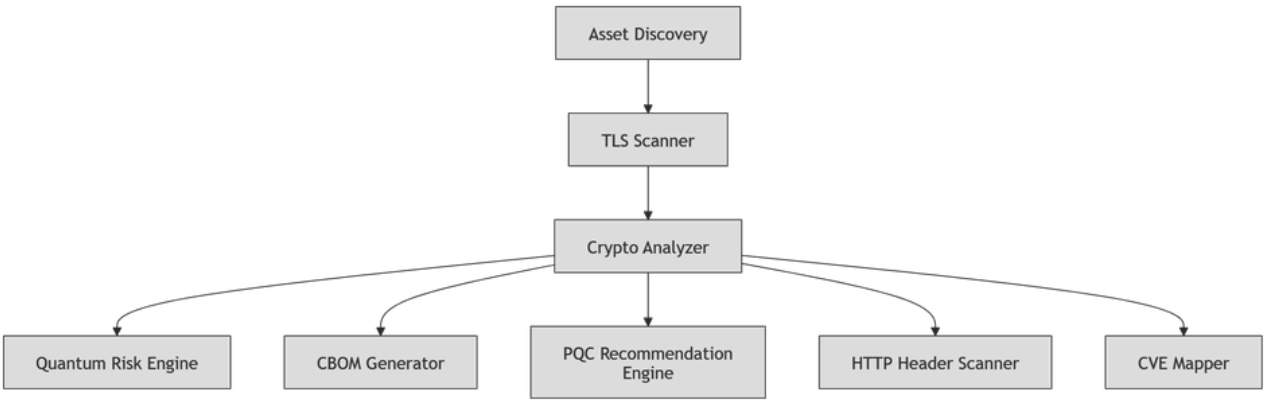
2.1 Product Perspective

QuantumShield is a modular cybersecurity platform that scans public-facing banking infrastructure for cryptographic vulnerabilities and evaluates their readiness for quantum-resistant security.

The system architecture consists of:



Scanning Pipeline Architecture:



2.2 Product Functions

QuantumShield provides the following major functions:

- Asset discovery using subdomain enumeration and network scanning
- TLS protocol and cipher suite detection
- Cryptographic algorithm classification
- Detection of insecure algorithms and protocols
- Generation of Cryptographic Bill of Materials (CBOM)
- Quantum risk scoring
- PQC migration recommendations

- Detection of known cryptographic attacks
- HTTP security header validation
- Dashboard visualization and reporting

2.3 User Classes and Characteristics

Examples:

- **Primary Users:** PNB Cybersecurity Team, PNB IT Division (Head Office), PNB System Administrators.
- **Secondary Users:** PNB Information Security Group (ISG), PNB Compliance Auditors. Users are expected to have technical knowledge of cryptography and networking.

User at	User Type	Menus for User
PNB / IIT Kanpur officials	Admin User	All administrative menus
PNB	Checker	Review and audit menus

2.4 Operating Environment

The operating environment for the 'QuantumShield' as listed below.

- Server system
- Operating system: Linux (Kali / Ubuntu)
- Database: MongoDB
- Platform: Web Browser
- Technology: Python, React, FastAPI
- API: RESTful APIs

2.5 Design and Implementation Constraints

1. Technical Constraints: - (For Deployment)

- **Network Configuration:** e.g. The application must support private (intranet) IP addressing. Appropriate firewalls and routing rules must be configured.

- **Hosting Environment:** e.g. Should deploy in intranet i.e. for intranet access.

2. Security Constraints

- **Access Control:** Ex: RBAC
- **Data Encryption:** Ex: HTTPS uses all the data transmitted over the internet.

3. Performance Constraints

- **Failover Mechanisms:** Ex: Ensure redundancy and failover mechanisms are in place for both environments to maintain availability.

4. User Interface Constraints

- **User Experience Consistency:** Ex: Maintaining the consistent design and navigation elements across both environments to minimize confusion for users switching between them.

Examples:

- Must submit with NIST PQC standards.
- Must operate only on public-facing applications.
- Must not disrupt live banking services.
- Must generate reports in machine-readable formats (JSON, XML, CSV).

2.6 Assumptions and Dependencies

Assumptions:

- **Standard Browser Support:** e.g. It is assumed that end users will be accessing the application using HTML5-compliant browser such as Google Chrome.
- **Examples:** Assumes TLS-based communication is used in all public-facing applications. Assumes internet connectivity for scanning endpoints.

Dependencies:

- **Database System:** Ex: The application depends on MongoDB database for data storage. Any maintenance, downtime, or performance issues with the database will directly effect on the application's functionality.
- **Examples:** Depends on NIST PQC algorithms that are `sbeing standardized and available.

3. Specific Requirements

3.1 Functional Requirements

The system shall:

- FR1 – Discover public-facing assets using domain scanning tools.
- FR2 – Identify supported TLS protocol versions.
- FR3 – Enumerate all supported cipher suites.
- FR4 – Detect certificate chain and expiration information.
- FR5 – Classify cryptographic algorithms according to risk level.
- FR6 – Detect known cryptographic vulnerabilities.
- FR7 – Generate Cryptographic Bill of Materials (CBOM).
- FR8 – Calculate a Quantum Readiness Score.
- FR9 – Provide migration recommendations to PQC algorithms.
- FR10 – Display results via a web dashboard.

3.2 External Interface Requirements

3.2.1 User Interfaces

The system will be used to provide dashboard which gives access to users in:

- Initiate scans
- View scan results
- Monitor cryptographic risk scores
- Download CBOM reports

The dashboard will be accessible via modern browsers such as:

- Google Chrome
- Microsoft Edge
- Firefox

3.2.2 Hardware Interfaces

The system requires:

- Server infrastructure for backend processing
- Network interface for scanning external systems
- Storage infrastructure for storing scan results

3.2.3 Software / Communication Interfaces

The backend interacts with:

- MongoDB database
- React frontend via REST APIs
- External scanning tools

APIs integrated are:

- POST /api/v1/scan
- GET /api/v1/results/{domain}
- GET /api/v1/cbom/{domain}
- GET /api/v1/quantum-score/{domain}

3.3 System Features

Key features of Quantum Shield involves:

- Asset discovery engine
- TLS protocol analysis
- Cipher suite detection
- Certificate chain validation
- Quantum readiness scoring
- Cryptographic Bill of Materials
- PQC migration recommendations
- CVE vulnerability detection
- HTTP security header validation
- Centralized dashboard monitoring

3.4 Non-Functional Requirements

3.4.1 Performance Requirements

- The system should scan a domain within 2 minutes.
- The scanning pipeline should operate asynchronously.
- API response times should be under 500 ms.

3.4.2 Software Quality Attributes

Attribute	Description
Reliability	System should operate continuously without crashes

Scalability	Must support scanning of multiple domains
Maintainability	Modular architecture for easy updates
Usability	Dashboard must be intuitive

3.4.3 Other Non-Functional Requirements

- Logs must be maintained for all scan operations.
- Reports should be exportable in JSON, CSV, and PDF formats.
- System must support future PQC algorithm updates.

4. Technological Requirements

4.1 Technologies Used

Layer	Technology
Backend	FastAPI
Language	Python 3.11
Frontend	React
Database	MongoDB
Scanning Tools	Nmap, Subfinder
Cryptography Analysis	Python ssl, cryptography

4.2 IDE

Development tools are:

- Visual Studio Code
- PyCharm
- GitHub for version control

4.3 Database Management

MongoDB is being used to store:

- Scan results
- Cryptographic inventories
- CBOM reports
- Risk scores

5. Security Requirements

Security mechanisms implement by the system are:

Access Control

- Role-Based Access Control (RBAC)
- Authentication for admin users

Data Security

- Secure storage of scan results
- Encryption for data in transit

Logging and Auditing

- All scanning activities must be logged
- Logs must include timestamp, user ID, and scan target

Compliance

- Must comply with NIST Post-Quantum Cryptography standards
- Must align with CERT-IN CBOM reporting guidelines

System Recovery

- Backup and recovery mechanisms for the database
- Failover mechanisms used to maintain service availability

Annexure-A (CERT-IN CBOM elements)

Table 9 : Minimum Elements pertaining to Cryptographic Asset

Cryptographic Asset Type	Element	Description
Algorithms	Name	The name of the cryptographic algorithm or asset. For example, "AES-128-GCM" refers to the AES algorithm with a 128-bit key in Galois/Counter Mode (GCM).
	Asset Type	Specifies the type of cryptographic asset. For algorithms, the asset type is "algorithm".
	Primitive	Describes the cryptographic primitive. For "SHA512withRSA", the primitive is "signature" as it's used for digital signing.
	Mode	The operational mode used by the algorithm. For example, "gcm" refers to the Galois/Counter Mode used with AES encryption.

Cryptographic Asset Type	Element	Description
	Crypto Functions	The cryptographic functions supported by the asset. For example, the functions in the case of "AES-128-GCM" are key generation, encryption, decryption, and authentication tag generation.
	Classical security level	The classical security level represents the strength of the cryptographic asset in terms of its resistance to attacks using classical (non-quantum) methods. For AES-128, it's 128 bits.
	OID	The Object Identifier (OID) is a globally unique identifier used to refer to the algorithm. It helps in distinguishing algorithms across different systems. For example, "2.16.840.1.101.3.4.1.6" for AES-128-GCM, "1.2.840.113549.1.1.13" for SHA512withRSA
	List	Lists the cryptographic algorithms employed by the quantum device or system, allowing for an assessment of its security capabilities, especially in the context of post-quantum encryption standards.
Keys	Name	The name of the key, which is a unique identifier for the key used in cryptographic operations.
	Asset Type	Defines the type of cryptographic asset. For keys, the asset type is typically "key".
	id	A unique identifier for the key, such as a key ID or reference number.
	state	The state of the key, such as whether it is active, revoked, or expired.
	size	The size of the key, typically measured in bits. For example, a 128-bit key or a 2048-bit RSA key.
	Creation Date	The date when the key was created.
	Activation Date	The date when the key became operational or was first used.
Protocols	Name	The name of the cryptographic protocol, such as TLS, IPsec, or SSH

Cryptographic Asset Type	Element	Description
	Asset Type	Defines the type of cryptographic asset. In this case, it would be a "protocol"
	Version	The version of the protocol used, such as TLS 1.2 or TLS 1.3.
	Cipher Suites	The set of cryptographic algorithms and parameters supported by the protocol for tasks like encryption, key exchange, and integrity checking.
	OID	The Object Identifier (OID) associated with the protocol, identifying its unique specifications.
Certificates	Name	The name of the certificate, typically referring to its subject or the entity it represents (e.g., a website).
	Asset Type	Defines the type of cryptographic asset. For certificates, the asset type is "certificate".
	Subject Name	This refers to the Distinguished Name (DN) of the entity that the certificate represents. It typically contains information about the organization, domain name
	Issuer Name	The issuer is the Certificate Authority (CA) that issued and signed the certificate. This field contains the DN of the CA that verified and issued the certificate.
	Not Valid Before	This specifies the date and time from which the certificate is valid.
	Not Valid After	This specifies the expiration date and time of the certificate. The certificate becomes invalid after this timestamp.
	Signature Algorithm Reference	This refers to the cryptographic algorithm used to sign the certificate. It provides a reference to the algorithm and its OID (Object Identifier).
	Subject Public Key Reference	This points to the public key used by the subject (the entity being identified in the certificate). It provides a reference to the key's details, including the algorithm.
	Certificate Format	Specifies the format of the certificate. Common formats include X.509, which is the most widely used format for certificates.

Cryptographic Asset Type	Element	Description
	Certificate Extension	This refers to the file extension associated with the certificate. It is commonly .crt for certificates in the X.509 format.